

Task 2: Synthesis & Gate-Level Simulation

Title: RTL Synthesis and Gate-Level Netlist Generation using Yosys

Objective

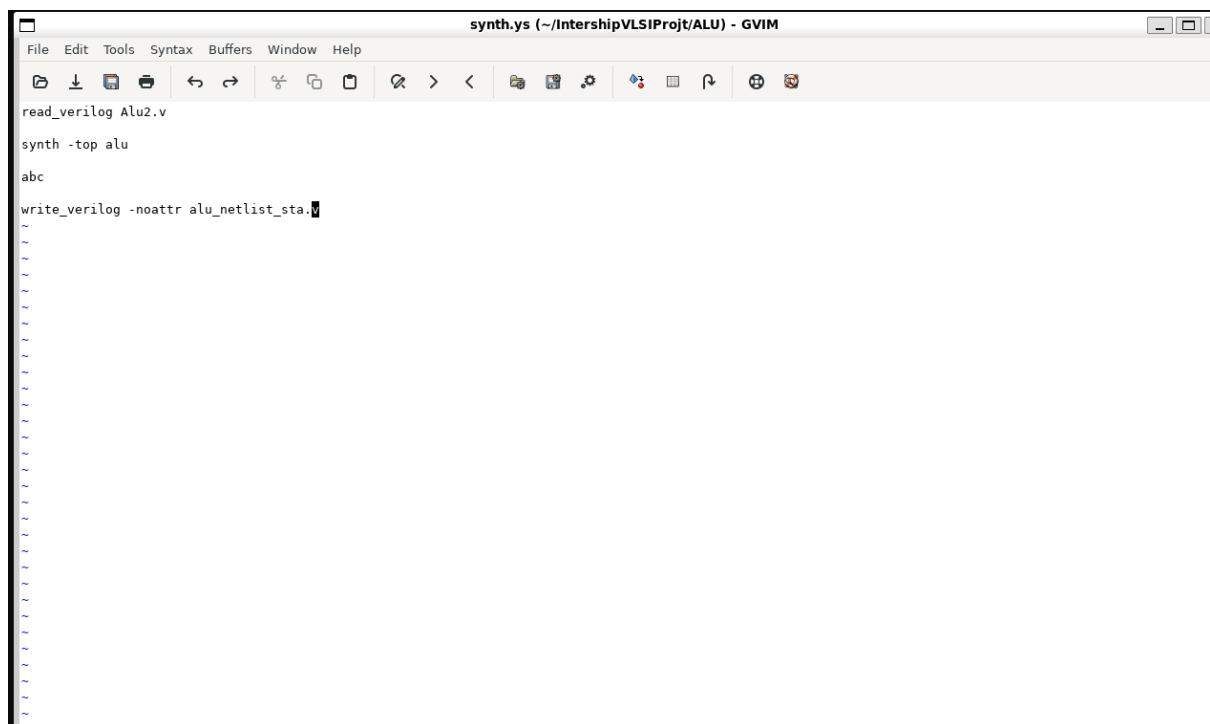
Synthesize the RTL ALU design into a gate-level netlist using Yosys and analyze synthesis results.

Tool Used

- Yosys 0.33
- ABC Logic Optimizer

Output Screenshot :

1: **synth.js** output → Screenshot



```
synth.js (~/IntershipVLSIProjt/ALU) - GVIM
File Edit Tools Syntax Buffers Window Help
read_verilog Alu2.v
synth -top alu
abc
write_verilog -noattr alu_netlist_sta.~
```

2: **Yosys synthesis output** → Screenshot

- **ABC RESULTS** section → Screenshot

```
3.1.2. Re-integrating ABC results.
ABC RESULTS:      NOR cells:      3
ABC RESULTS:      ANDNOT cells:    6
ABC RESULTS:      XNOR cells:      2
ABC RESULTS:      XOR cells:       7
ABC RESULTS:      OR cells:        23
ABC RESULTS:      ORNOT cells:     17
ABC RESULTS:      AND cells:       66
ABC RESULTS:      NAND cells:      82
ABC RESULTS:      internal signals: 231
ABC RESULTS:      input signals:   19
ABC RESULTS:      output signals:   9
Removing temp directory.

4. Executing Verilog backend.

4.1. Executing BMUXMAP pass.

4.2. Executing DEMUXMAP pass.
Dumping module '\alu'.

End of script. Logfile hash: 1af8f5294d, CPU: user 0.23s system 0.05s, MEM: 13.75 MB peak
Yosys 0.33 (git sha1 2584903a060)
Time spent: 51% 2x abc (0 sec), 11% 16x opt_expr (0 sec), ...
govind@LAPTOP-HQSI9PA0:~/IntershipVLSIProjt/ALU$
```

3: head -50 alu_netlist.v output → Screenshot

```
govind@LAPTOP-HQSI9PA0:~/IntershipVLSIProjt/ALU$ head -50 alu_netlist.v
/* Generated by Yosys 0.33 (git sha1 2584903a060) */

(* top = 1 *)
(* src = "Alu2.v:1.1-48.10" *)
module alu(A, B, opcode, result, carry);
  wire _000_;
  wire _001_;
  wire _002_;
  wire _003_;
  wire _004_;
  wire _005_;
  wire _006_;
  wire _007_;
  wire _008_;
  wire _009_;
  wire _010_;
  wire _011_;
  wire _012_;
  wire _013_;
  wire _014_;
  wire _015_;
  wire _016_;
  wire _017_;
  wire _018_;
  wire _019_;
  wire _020_;
  wire _021_;
  wire _022_;
  wire _023_;
  wire _024_;
  wire _025_;
  wire _026_;
  wire _027_;
  wire _028_;
  wire _029_;
  wire _030_;
  wire _031_;
  wire _032_;
  wire _033_;
  wire _034_;
  wire _035_;
  wire _036_;
  wire _037_;
  wire _038_;
  wire _039_;
  wire _040_;
  wire _041_;
  wire _042_;
  wire _043_;
  wire _044_;
govind@LAPTOP-HQSI9PA0:~/IntershipVLSIProjt/ALU$ |
```

4: Gate-level simulation output → Screenshot

```

Time spent: 32% 2x abc (0 sec), 10% 10x opt_exp (0 sec), ...
govind@LAPTOP-HQSI9PA0:~/IntershipVLSIProjt/ALU$ iverilog alu_netlist.v Alu2_tb.v -o gate_sim
govind@LAPTOP-HQSI9PA0:~/IntershipVLSIProjt/ALU$ vvp gate_sim
VCD info: dumpfile alu.vcd opened for output.
t=0 A=0a B=05 opcode=000 result=0f
t=10000 A=14 B=08 opcode=001 result=0c
t=20000 A=aa B=cc opcode=010 result=88
t=30000 A=aa B=cc opcode=011 result=ee
t=40000 A=aa B=cc opcode=100 result=66
t=50000 A=aa B=cc opcode=101 result=55
t=60000 A=aa B=cc opcode=110 result=54
t=70000 A=aa B=cc opcode=111 result=55
t=80000 A=24 B=81 opcode=001 result=a3
t=90000 A=63 B=0d opcode=101 result=9c
t=100000 A=65 B=12 opcode=001 result=53
t=110000 A=0d B=76 opcode=101 result=f2
t=120000 A=ed B=8c opcode=001 result=61
t=130000 A=c6 B=c5 opcode=010 result=c4
t=140000 A=e5 B=77 opcode=010 result=65
t=150000 A=8f B=f2 opcode=110 result=1e
t=160000 A=e8 B=c5 opcode=100 result=2d
t=170000 A=bd B=2d opcode=101 result=42
t=180000 A=63 B=0a opcode=000 result=6d
t=190000 A=20 B=aa opcode=101 result=df
t=200000 A=96 B=13 opcode=101 result=69
t=210000 A=53 B=6b opcode=101 result=ac
t=220000 A=02 B=ae opcode=101 result=fd
t=230000 A=cf B=23 opcode=010 result=03
t=240000 A=ca B=3c opcode=010 result=08
t=250000 A=8a B=41 opcode=000 result=cb
t=260000 A=78 B=89 opcode=011 result=f9
t=270000 A=b6 B=c6 opcode=110 result=6c
t=280000 A=bc B=2a opcode=011 result=be
t=290000 A=71 B=85 opcode=111 result=38
t=300000 A=3b B=3a opcode=110 result=76
t=310000 A=15 B=f1 opcode=001 result=24
t=320000 A=62 B=4c opcode=111 result=31
t=330000 A=8f B=f8 opcode=111 result=47
t=340000 A=9f B=5c opcode=011 result=df
t=350000 A=89 B=49 opcode=000 result=d2
t=360000 A=d7 B=51 opcode=110 result=ae
t=370000 A=0c B=c2 opcode=000 result=ce
t=380000 A=77 B=3d opcode=010 result=35
t=390000 A=7e B=6d opcode=001 result=11
t=400000 A=1f B=d3 opcode=101 result=e0
t=410000 A=78 B=5b opcode=001 result=1d
t=420000 A=3f B=2a opcode=000 result=69
t=430000 A=86 B=8e opcode=100 result=08
t=440000 A=fa B=26 opcode=011 result=fe
t=450000 A=a3 B=2f opcode=011 result=af
t=460000 A=5f B=44 opcode=111 result=2f
t=470000 A=cb B=e6 opcode=010 result=c2
t=480000 A=29 B=ed opcode=010 result=29
t=490000 A=65 B=b5 opcode=111 result=32
t=500000 A=79 B=44 opcode=000 result=bd
t=510000 A=2a B=ab opcode=110 result=54
t=520000 A=dc B=9a opcode=101 result=23
t=530000 A=c3 B=56 opcode=110 result=86
t=540000 A=67 B=0a opcode=110 result=ce
t=550000 A=38 B=79 opcode=000 result=b1
t=560000 A=94 B=93 opcode=100 result=07
t=570000 A=59 B=db opcode=101 result=a6
Alu2_tb.v:70: $finish called at 580000 (1ps)
govind@LAPTOP-HQSI9PA0:~/IntershipVLSIProjt/ALU$ |

```

Task 2 : all commands

- 1) `cd ~/IntershipVLSIProjt/ALU`
- 2) `gvim synth.vs`
- 3) `yosys -s synth.vs`
- 4) `grep "ABC RESULTS" -A 15 synthesis.log`
- 5) `head -50 alu_netlist.v`
- 6) `iverilog alu_netlist.v Alu2_tb.v -o gate_sim`
- 7) `vvp gate_sim`